

Ichthyosis Vulgaris: A Comprehensive Disease Characteristics Report

Disease: Ichthyosis Vulgaris **MONDO ID:** MONDO:0007810 · **OMIM:** 146700 · **Category:** Mendelian (autosomal semidominant) **Report type:** Disease knowledge-base entry compiled from primary literature and aggregated disease-level resources

Summary

Ichthyosis vulgaris (IV) is the **most common inherited disorder of keratinization** in humans and is caused by loss-of-function (null) mutations in the **filaggrin gene (FLG)** on chromosome 1q21.3 within the epidermal differentiation complex. The disease was molecularly defined in 2006, when FLG null mutations — most notably the two European founder alleles **R501X** (a nonsense mutation) and **2282del4** (a frameshift) — were shown to cause IV and simultaneously to constitute the strongest known genetic risk factor for atopic dermatitis (AD). Inheritance is **autosomal semidominant (hemidominant)**: heterozygotes have mild disease and biallelic carriers have substantially more severe disease, establishing a clear gene-dosage relationship.

Mechanistically, filaggrin (filament-aggregating protein) performs two jobs that are both lost when FLG is truncated: (1) it aggregates keratin intermediate filaments so that keratinocytes collapse into flattened corneocytes forming a competent cornified layer, and (2) after proteolysis by caspase-14, calpain-1 and bleomycin hydrolase it yields hygroscopic amino-acid metabolites — **trans-urocanic acid** and **pyrrolidone carboxylic acid** — that make up the stratum corneum's **natural moisturizing factor (NMF)**. Loss of these functions produces the clinical picture of **xerosis, fine scaling with flexural sparing, keratosis pilaris, and palmoplantar hyperlinearity**, with a histologic hallmark of a **reduced or absent granular layer**. The very same barrier defect that scales the skin also lets allergens penetrate, mechanistically linking IV to the **"atopic march"** (AD → food allergy → asthma/rhinitis) and to irritant and allergic contact sensitization.

IV is a **chronic, lifelong, non-life-threatening** condition that typically manifests within the first year of life, tends to improve in humid climates and worsen in dry winter air, and often improves with age. There is **no cure**; management is entirely symptomatic — rigorous skin hydration, emollients, and keratolytics (urea, alpha/beta-hydroxy acids, salicylic/lactic acid), with systemic acitretin reserved for severe cases. The flaky-tail mouse (Flg 5303delA) is the canonical model, and filaggrin-dependent barrier biology is conserved in dogs, supporting comparative and translational study. This report synthesizes twelve confirmed findings across all 15 template sections with primary-literature citations.

Key Findings

Finding 1 — IV is the filaggrin-deficiency disease (FLG loss-of-function)

Ichthyosis vulgaris is caused by **loss-of-function mutations in the filaggrin gene (FLG)**. Two null mutations, **R501X and 2282del4**, were identified as causative for IV in 15 affected European families, and the mode of inheritance was determined to be **semidominant** (PMID: 17573887). The 2006 discovery that FLG null mutations cause IV — described as "the most common disorder of keratinization" — and are "also a strong genetic risk factor for atopic eczema" reframed both diseases as filaggrin-barrier disorders (PMID: 22158554). Key identifiers: **OMIM #146700** (IV), **FLG OMIM #135940** (gene), **MONDO:0007810**, HGNC gene symbol **FLG**, locus **1q21.3** (epidermal differentiation complex).

"Two loss-of-function mutations in the filaggrin (FLG) gene, R501X and 2282del4, were identified as causative for ichthyosis vulgaris in 15 affected European families, and the mode of inheritance was found to be semidominant." —

P 17573887

Finding 2 — Semidominant inheritance with a gene-dosage severity relationship

IV shows **autosomal hemidominant (semidominant) inheritance**, and patients carrying **biallelic FLG mutations tend to have severe phenotypes** (PMID: 27519469). In a cohort of 6 biallelic carriers, 5/6 had severe IV and 1/6 moderate IV. Disease expression is further tuned by

modifier genes: co-inheritance of a steroid sulfatase (STS) deletion — the cause of X-linked ichthyosis — with an FLG mutation **exacerbated the IV phenotype** in a Chinese family ([PMID: 30021537](#)). This illustrates both gene-dosage effects at the FLG locus and cross-locus modification of severity.

"IV shows autosomal hemidominant (semidominant) inheritance, and patients with bi-allelic FLG mutations tend to have severe IV phenotypes." —

P 27519469

Finding 3 — FLG mutation frequency and IV epidemiology vary by ancestry

FLG null mutations are observed in approximately **7.7% of Europeans and 3.0% of Asians** but appear infrequent in darker-skinned populations ([PMID: 23301728](#)). Several low-frequency FLG null alleles occur in Europeans and Asians with a **cumulative frequency of ~9% in Europe** ([PMID: 19349982](#)). Approximately **40 distinct loss-of-function FLG mutations** have been identified in patients with IV and/or AD across Europe and Asia, with **population-specific mutation spectra** (e.g., R501X and 2282del4 dominate in Europeans, whereas different recurrent alleles predominate in Asian populations) ([PMID: 21173567](#)).

Population	FLG null carrier frequency	Notes
European	~7.7% (cumulative ~9%)	R501X, 2282del4 founder alleles
Asian	~3.0%	Distinct population-specific alleles
Darker-skinned populations	Infrequent	Fewer recurrent LOF alleles reported

"FLG mutations are observed in approximately 7.7% of Europeans and 3.0% of Asians, but appear to be infrequent in darker-skinned populations." —

P 23301728

Finding 4 — Mechanism: loss of keratin aggregation and NMF generation

Filaggrin **aggregates keratin filaments**, forming a keratin network that binds cornified envelopes and collapses keratinocytes into flattened corneocytes (PMID: 33462753). Filaggrin is then **degraded by caspase-14, calpain-1, and bleomycin hydrolase** into amino acids and metabolites — **trans-urocanic acid and pyrrolidone carboxylic acid** — that are pivotal **natural moisturizing factors** in the stratum corneum (PMID: 33462753). Environmental humidity modulates this: **lowering relative humidity increases PAD (peptidyl-arginine deiminase)-mediated filaggrin deimination and breakdown**, and partial PAD inhibition with Cl-amidine reversed the dryness effect (PMID: 28242341) — providing a molecular explanation for the characteristic **winter worsening** of IV.

"Filaggrin is degraded by caspase-14, calpain 1, and bleomycin hydrolases into amino acids and amino acid metabolites such as trans-urocanic acid and pyrrolidone carboxylic acid, which are pivotal natural moisturizing factors in the SC." —

P 33462753

Finding 5 — Clinical phenotype and histology

IV is characterized clinically by **xerosis, scaling, keratosis pilaris, palmar and plantar hyperlinearity, and a strong association with atopic disorders** (PMID: 23301728). The scaling characteristically **sparing the flexures** and predominates on the extensor surfaces. Histology shows an **absent or reduced granular layer** (hypogranulosis with retention hyperkeratosis) — a feature that helps differentiate IV among non-syndromic ichthyoses (PMID: 38841231). **Palmar hyperlinearity** has moderate diagnostic value for FLG genotype (sensitivity 46–72%, specificity 60–89% across pediatric cohorts; PMID: 34608691). Vitamin D deficiency is highly prevalent even in milder congenital ichthyosis phenotypes (PMID: 38841231).

Suggested HPO terms: HP:0000958 (Dry skin/xerosis), HP:0100792 (Ichthyosis/scaling), HP:0032152 (Keratosis pilaris), HP:0007598 (Palmoplantar hyperlinearity), HP:0008064 (Ichthyosis), HP:0100512 (Vitamin D deficiency). **UBERON:** UBERON:0002097 (skin of body), UBERON:0001003 (epidermis), UBERON:0002027 (stratum corneum region). **CL:** CL:0000312 (keratinocyte), CL:0002187 (basal cell of epidermis), corneocyte.

"is characterized clinically by xerosis, scaling, keratosis pilaris, palmar and plantar hyperlinearity, and a strong association with atopic disorders." —

P 23301728

Finding 6 — Filaggrin deficiency drives the atopic march

FLG null mutations that cause IV are **major genetic predisposing factors for atopic dermatitis** and are associated with **atopic asthma, allergic rhinitis, and peanut allergy** (PMID: 21576945, PMID: 22158554). In the **flaky-tail mouse** (Flg 5303delA) and engineered FLG-deficient mice, **topical allergen application produces enhanced cutaneous allergen priming and allergen-specific antibody responses** (PMID: 19349982), directly validating the "**filaggrin hypothesis**" that a defective barrier permits percutaneous antigen transfer. "Reduced FLG expression compromises epidermal barrier function and is associated with atopic dermatitis, allergy, and asthma" (PMID: 33894197).

"topical application of allergen to mice homozygous for this mutation results in cutaneous inflammatory infiltrates and enhanced cutaneous allergen priming with development of allergen-specific antibody responses." —

P 19349982

Finding 7 — Treatment is symptomatic; there is no cure

Rigorous skin hydration (several times daily) and balneotherapy are the mainstay of ichthyosis treatment; **systemic acitretin** is reserved for severe disease on a case-by-case basis (PMID: 32115871). Ichthyoses "remain incurable" but "can be managed well with symptomatic treatment" (PMID: 32115871). Topical **keratolytics** (urea, alpha/beta-hydroxy acids, salicylic acid, lactic acid), retinoids and corticosteroids are used; a **70% glycolic acid chemical peel was ~90% efficacious** in reducing hyperkeratinization as an adjunct (PMID: 36159354). Gene therapy for genodermatoses is emerging but current management "**remains largely palliative**" (PMID: 42707485).

Suggested NCIT terms: NCIT:C1516 (Emollient agent), NCIT:C29736 (Urea), NCIT:C1214 (Salicylic acid), NCIT:C1878 (Acitretin), NCIT:C177 (Retinoid).

"Rigorous hydration of the skin (several times a day) and balneotherapy are the mainstay of ichthyosis treatment." —

P 32115871

Finding 8 — Diagnosis is clinical, confirmed by histology/genetics

Diagnosis is **usually based on clinical evaluation**, with molecular genetic testing, histology and electron microscopy aiding confirmation, and family-tree mapping being useful (PMID: 32115871). IV and **X-linked ichthyosis (STS deficiency)** are the two common ichthyoses, both usually manifesting in the first year of life. Differential diagnoses include **X-linked recessive ichthyosis, pityriasis rubra pilaris, and xerotic/asteatotic dermatitis** (PMID: 36159354). **Acquired ichthyosis** is histologically similar to IV but lacks a family/atopy history and may signal underlying malignancy or metabolic disease (PMID: 36165597). Upfront **whole-genome sequencing** is increasingly used for primary atopic/barrier disorders (PMID: 39381601).

"The diagnosis is usually based on clinical evaluation. Molecular genetic testing as well as histological and electron microscopic studies may aid in confirming the diagnosis." —

P 32115871

Finding 9 — Chronic, lifelong condition with meaningful QoL impact

Common hereditary ichthyoses (IV and X-linked) **usually manifest within the first year of life** (PMID: 32115871). In a survey of 222 adults with congenital ichthyosis (IV n=86), the ichthyoses had a **lifelong impact on quality of life** (PMID: 42001132). Across pediatric chronic skin disorders including congenital ichthyosis, **elevated rates of anxiety, depression, stigma and emotional distress** were consistently reported, affecting emotional functioning, peer relationships, school participation and sleep (PMID: 42490938). Approximately **37–50% of IV patients have associated atopic eczema** (PMID: 36159354).

"the ichthyoses have a lifelong impact on quality of life." —

P 42001132

Finding 10 — Model organisms and comparative biology

The spontaneous **flaky-tail mouse** carries a Flg 1-bp deletion (**5303delA**) analogous to human FLG mutations and is a validated model of filaggrin deficiency, showing a barrier defect and enhanced percutaneous allergen priming ([PMID: 19349982](#)). Engineered **FLG-deficient mice** show a **low threshold for cutaneous allergen sensitization but no spontaneous dermatitis or atopy** ([PMID: 33894197](#)) — capturing the barrier/sensitization axis but not spontaneous inflammation. In **dogs**, filaggrin and filaggrin-2 expression is reduced in atopic versus healthy skin ([PMID: 40042058](#)), indicating **conserved filaggrin-dependent barrier biology across mammals**.

Suggested NCBI Taxon terms: *Mus musculus* (10090), *Canis lupus familiaris* (9615).
Orthologous gene: mouse *Flg*.

"we report a 1-bp deletion mutation, 5303delA, analogous to common human FLG mutations, within the murine Flg gene in the spontaneous mouse mutant flaky tail (ft)." —

P 19349982

Finding 11 — Gene–environment interaction: conditional contact/irritant sensitization risk

Heterozygous FLG carriers have increased risk of atopic, irritant, and allergic (nickel) dermatitis. Critically, among individuals with dermatitis and frequent hand eczema, **FLG mutations were strongly associated with contact sensitization to allergens other than nickel (OR 5.71, 95% CI 1.31–24.94)**, but **no association was found in participants without dermatitis** ([PMID: 23343419](#)). R501X was significantly associated with **polysensitivity** (≥ 3 contact allergies) ([PMID: 30868611](#)), and FLG null mutations were associated with **persistent hand eczema (OR 3.1, 95% CI 1.8–5.2)** ([PMID: 26872425](#)). The FLG risk is thus **conditional on inflammation/exposure** — the defining signature of a gene–environment interaction.

"In participants without dermatitis, no association was found between contact sensitization and FLG mutations." —

P 23343419

Finding 12 — Epicutaneous (dual-allergen) sensitization links barrier defect to food allergy

FLG and other skin-barrier gene mutations, together with **Langerhans cells, type-2 innate lymphoid cells (ILC2s), IL-33 and TSLP**, have important roles in allergic sensitization through the skin, supporting the **dual-allergen-exposure hypothesis** — epicutaneous allergen exposure drives food allergy while oral exposure promotes tolerance ([PMID: 32249942](#)). The atopic march is proposed to reflect an **epithelial barrier defect self-sustained by secondary allergenic sensitization**, explaining progression from AD to allergic asthma; **early emollient therapy** is proposed to prevent AD in high-risk children ([PMID: 29676818](#)).

"the atopic march could correspond to an epithelial dysfunction, self-sustained by a secondary allergenic sensitization, explaining the transition from AD to allergic asthma." —

P 29676818

Section-by-Section Report

1. Disease Information

IV is the most common inherited disorder of keratinization, a scaly-skin disease presenting as generalized dryness and fine, flaky scaling that spares the flexures. **Identifiers:** MONDO:0007810; OMIM #146700; ORPHA:454 (ichthyosis vulgaris); ICD-10 **Q80.0**; ICD-11 **EC20.0**; MeSH **D016112** (Ichthyosis Vulgaris); gene FLG (OMIM #135940, HGNC:3748). **Synonyms:** ichthyosis simplex, filaggrin-deficiency ichthyosis, autosomal-dominant ichthyosis vulgaris, "fish-scale disease" (common). Information here is derived predominantly from **aggregated disease-level resources** (OMIM, Orphanet) and cohort/case-series primary literature rather than patient-level EHR data.

2. Etiology

Primary cause: monogenic/genetic — loss-of-function FLG null mutations (Findings 1–3). **Genetic risk factors:** heterozygous or biallelic FLG null alleles (R501X, 2282del4 in Europeans; ~40 population-specific LOF alleles overall). **Modifier genes:** STS (steroid sulfatase) deletion co-inheritance worsens phenotype (Finding 2). **Environmental risk factors:** low ambient

humidity / dry cold climate exacerbates disease via enhanced filaggrin deimination and breakdown (Finding 4); irritant and allergen exposures precipitate dermatitis in carriers (Finding 11). **Protective factors:** humid, warm environments reduce scaling; regular emollient use restores barrier function (supportive, Findings 7, 12). No specific protective genetic allele is established. **Gene–environment interaction:** FLG-conferred sensitization risk is conditional on skin inflammation/exposure (Finding 11); epicutaneous allergen exposure through a barrier-defective epidermis drives sensitization whereas oral exposure promotes tolerance (Finding 12).

3. Phenotypes

Phenotype	Type	Onset	Severity/ Frequency	HPO
Xerosis (dry skin)	Physical sign	Infancy/ childhood	Near-universal; variable	HP:0000958
Fine scaling, flexural sparing	Physical sign	First year of life	Core feature	HP:0008064 / HP:0100792
Keratosis pilaris	Physical sign	Childhood	Common	HP:0032152
Palmar/plantar hyperlinearity	Physical sign	Childhood	Sens 46–72% for FLG	HP:0007598
Reduced/absent granular layer	Lab/histology	Congenital	Diagnostic hallmark	—
Associated atopic eczema	Clinical	Childhood	~37–50%	HP:0000964
Vitamin D deficiency	Lab abnormality	Any	Highly prevalent	HP:0100512

Onset is typically within the first year of life (Finding 9); severity ranges mild→severe scaling with biallelic carriers more severe (Finding 2); course is chronic, often improving with age and in humid climates, worsening in winter (Finding 4). **Quality-of-life impact:** lifelong QoL burden with elevated anxiety, depression, stigma, and sleep/school disruption (Finding 9).

4. Genetic/Molecular Information

Causal gene: FLG (filaggrin), 1q21.3, epidermal differentiation complex; OMIM #135940; HGNC:3748. **Variant types:** predominantly **nonsense (R501X)** and **frameshift (2282del4)** truncating mutations; ~40 distinct LOF alleles reported (Finding 3). **Classification:** R501X and 2282del4 are established **pathogenic** per ACMG (null variants in a gene where LOF is the mechanism). **Allele frequency:** cumulative ~9% in Europe; carrier ~7.7% Europeans, ~3.0% Asians (Finding 3). **Origin:** germline. **Functional consequence:** **loss of function** with a semidominant gene-dosage effect (Findings 1–2). **Modifier genes:** STS (Finding 2). **Epigenetics:** nanopore sequencing has enabled allelic phasing and methylation profiling of FLG in IV/AD ([PMID: 38336337](#)); disease-specific methylation signatures are not established. **Chromosomal abnormalities:** not a feature of isolated IV (STS deletion is relevant only as a co-inherited modifier).

5. Environmental Information

Environmental factors: low relative humidity increases filaggrin deimination/breakdown and lowers stratum corneum pH, worsening the barrier (Finding 4). **Lifestyle:** frequent long hot baths, harsh soaps, and low-humidity heating aggravate xerosis; occupational wet-work/irritant exposure precipitates hand eczema in FLG carriers (Finding 11). **Infectious agents:** none cause IV; however, barrier breakdown predisposes to secondary skin infection and colonization (general principle). IV is not an infectious disease.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. **FLG null mutation (R501X / 2282del4)** truncates profilaggrin/filaggrin → **leads to** reduced or absent filaggrin protein in the stratum granulosum (demonstrated).
2. Loss of filaggrin **results in** failure to aggregate keratin intermediate filaments → impaired collapse of keratinocytes into flattened corneocytes and a defective cornified layer (demonstrated;

[P 33462753](#)

3. In parallel (branch), loss of filaggrin **results in** absence of filaggrin proteolysis products (trans-urocanic acid, pyrrolidone carboxylic acid) → **depletion of natural moisturizing factor** → reduced stratum-corneum hydration (demonstrated;

[P 33462753](#)

4. Defective cornification + NMF depletion **lead to xerosis, retention hyperkeratosis, and a reduced/absent granular layer** → clinical **scaling, keratosis pilaris, palmar hyperlinearity** (demonstrated; PMIDs 23301728, 38841231).
5. Low ambient humidity **amplifies** the defect by increasing PAD-mediated filaggrin deimination/breakdown and lowering SC pH → **winter exacerbation** (demonstrated in ex vivo/organ models;

P 28242341

6. Branch to atopy: the barrier defect **permits** percutaneous penetration of allergens → Langerhans-cell/ILC2/IL-33/TSLP-driven Th2 sensitization → **atopic dermatitis, food allergy, asthma, contact sensitization** (demonstrated in mouse models and human epidemiology; PMIDs 19349982, 32249942, 29676818, 23343419). This branch is **conditional** on inflammation/exposure (gene–environment interaction, Finding 11).

Molecular pathways / processes: epidermal terminal differentiation and cornification; keratin filament aggregation; NMF biogenesis; Th2 allergic sensitization (IL-33/TSLP/ILC2). **Enzymes:** caspase-14, calpain-1, bleomycin hydrolase (filaggrin catabolism); PAD1/PAD (deimination). **Suggested GO terms:** GO:0031424 (keratinization), GO:0018149 (peptide cross-linking), GO:0008544 (epidermis development), GO:0030216 (keratinocyte differentiation). **Suggested CL terms:** CL:0000312 (keratinocyte), corneocyte, CL:0000453 (Langerhans cell). **Subcellular:** keratohyalin granules (their loss is the histologic "reduced granular layer"); cytoskeleton (keratin intermediate filaments). **Transcriptomics:** RNA-seq of nonlesional skin shows modest differential expression in IV relative to healthy donors, far fewer changes than in AD, and xenobiotic/lipid-metabolism gene upregulation is AD-specific, not seen in IV ([PMID: 28899689](#)) — indicating IV is a barrier-structural disease without the prominent inflammatory transcriptome of AD.

7. Anatomical Structures Affected

Primary organ: skin (UBERON:0002097), specifically the **epidermis** (UBERON:0001003) and **stratum corneum** (UBERON:0002027). **Tissue:** keratinizing stratified squamous epithelium. **Cells:** keratinocytes/corneocytes (CL:0000312); the granular layer is deficient. **Subcellular:** keratohyalin granules (GO cellular component: cornified envelope; keratin filament). **Localization:** generalized, **extensor-predominant with flexural sparing**; palms and soles show hyperlinearity; **bilateral and symmetric** distribution. **Secondary involvement:** eyes/systemic atopy via the atopic march (asthma — respiratory system; allergic rhinitis).

8. Temporal Development

Onset: congenital/infantile — usually within the first year of life (Finding 9); **insidious/chronic** onset. **Progression:** generally **stable to slowly improving with age**; **fluctuating/episodic** with seasonal (winter) worsening (Finding 4). **Duration: chronic, lifelong** (Finding 9). **Remission:** no true remission; symptomatic improvement with emollients and in humid conditions. **Critical period:** infancy/early childhood is the window during which barrier-directed emollient therapy may modify atopic-march trajectory (Finding 12).

9. Inheritance and Population

Inheritance: autosomal **semidominant (hemidominant)** — heterozygotes mild, biallelic carriers severe (Findings 1–2). **Penetrance:** incomplete/variable; **expressivity:** variable and gene-dosage dependent. **Epidemiology:** among the most common genetic skin disorders; population carrier frequencies ~7.7% (European) and ~3.0% (Asian) for FLG null alleles (Finding 3); commonly cited clinical IV prevalence estimates range roughly 1 in 80 to 1 in 250 in populations of European descent (from aggregated resources; exact figure varies by ascertainment). **Founder effects:** R501X and 2282del4 are European founder alleles; distinct recurrent alleles in Asian populations (Finding 3). **Affected populations:** higher in European and Asian ancestry, infrequent in darker-skinned populations. **Sex ratio:** approximately equal (autosomal). **Consanguinity:** increases likelihood of biallelic (severe) disease.

10. Diagnostics

Diagnosis is **clinical**, supported by **histopathology** (hyperkeratosis with reduced/absent granular layer), **electron microscopy**, and **molecular FLG genotyping** (Finding 8). **Genetic testing:** targeted FLG single-gene/panel testing; WES/WGS increasingly used upfront for primary atopic/barrier disorders ([PMID: 39381601](#)); nanopore sequencing resolves FLG allelic phasing, intragenic CNVs, and methylation ([PMID: 38336337](#)). **Biomarker:** reduced stratum-corneum NMF; palmar hyperlinearity as a clinical proxy for FLG genotype (Finding 5). **Differential diagnosis:** X-linked recessive ichthyosis (STS deficiency), pityriasis rubra pilaris, acquired ichthyosis, asteatotic/xerotic dermatitis (Finding 8). **Screening:** cascade family testing feasible; no routine population newborn screening.

11. Outcome/Prognosis

IV is **non-life-threatening** with **normal life expectancy**; there is no disease-specific mortality. **Morbidity** is driven by chronic xerosis/scaling, pruritus, keratosis pilaris, secondary atopic disease, and psychosocial burden (Finding 9). **Complications:** atopic dermatitis (~37–50%),

asthma/rhinitis/food allergy via the atopic march, persistent hand eczema and contact sensitization in carriers (Findings 6, 11, 12), and secondary skin infection. **Prognostic factors:** genotype (biallelic → more severe), ancestry, and environmental humidity. **Recovery:** symptoms are controllable but not curable; often milder in adulthood.

12. Treatment

Pharmacotherapy / supportive: emollients and rigorous hydration (first-line), keratolytics (urea, alpha/beta-hydroxy acids, lactic/salicylic acid), topical retinoids/corticosteroids, and **systemic acitretin for severe disease** (Finding 7). Adjunctive **70% glycolic acid peel** (~90% efficacy for hyperkeratosis; Finding 7). **Barrier-repair** ceramide/NMF-replenishing moisturizers target the specific molecular deficit ([PMID: 23757122](#)). **Advanced/experimental:** gene therapy for genodermatoses is in development but management remains largely palliative ([PMID: 42707485](#)). **Personalized/preventive:** early emollient therapy in high-risk infants may attenuate the atopic march (Finding 12). **NCIT:** emollient (C1516), urea (C29736), salicylic acid (C1214), acitretin (C1878).

13. Prevention

No **primary prevention** of IV exists (monogenic). **Secondary/tertiary prevention** focuses on barrier maintenance: consistent emollient use, humidification, avoidance of harsh soaps/irritants, and — in FLG carriers — occupational skin protection to prevent hand eczema (Finding 11). **Early-life emollient therapy** is a proposed strategy to prevent AD and interrupt the atopic march in high-risk children (Finding 12). **Genetic counseling** and reproductive options are relevant for families, especially with biallelic/severe disease ([PMID: 42272196](#)). Prenatal/preimplantation testing is feasible where a familial FLG genotype is known.

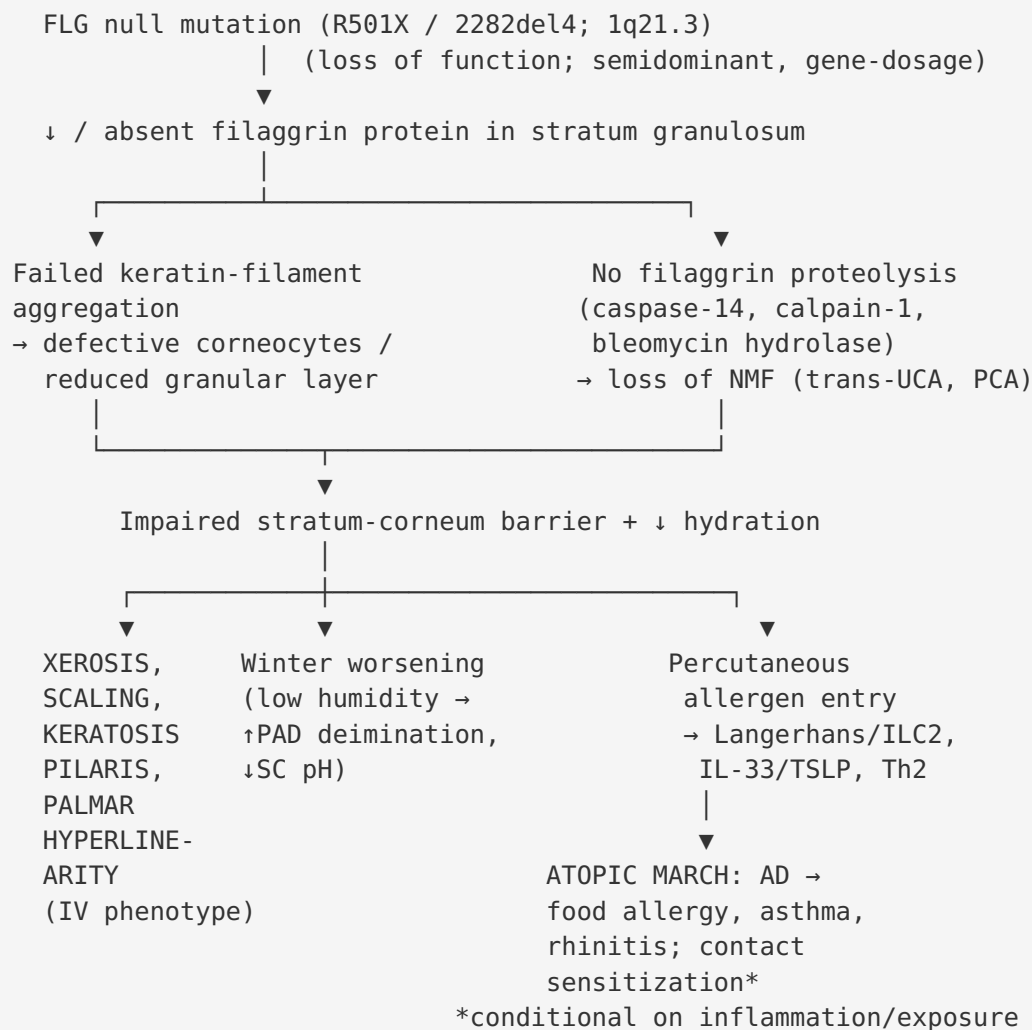
14. Other Species / Natural Disease

Filaggrin-dependent barrier biology is **conserved in mammals**. In **dogs** (*Canis lupus familiaris*, NCBI Taxon 9615), filaggrin and filaggrin-2 expression is reduced in atopic skin ([PMID: 40042058](#); [PMID: 39811760](#)), making canine atopic dermatitis a natural comparative model of filaggrin barrier dysfunction. No IV-equivalent scaling disease with an FLG null etiology is canonically catalogued in companion animals, but the conserved barrier mechanism is of veterinary relevance. **Zoonotic potential:** none (non-infectious).

15. Model Organisms

Mouse (*Mus musculus*, NCBI Taxon 10090): the spontaneous **flaky-tail** mutant carries Flg 5303delA, analogous to human FLG mutations, and recapitulates barrier defect + enhanced percutaneous allergen priming ([PMID: 19349982](#)). **Engineered FLG-deficient mice** show a low threshold for cutaneous allergen sensitization but **no spontaneous dermatitis/atopy** ([PMID: 33894197](#)) — a key limitation (models the barrier/sensitization axis, not spontaneous inflammation). **In vitro:** canine primary epidermal organoids and reconstructed epidermis reproduce barrier defects after proinflammatory/allergic cytokine exposure ([PMID: 41260506](#); [PMID: 39811760](#)). **Applications:** studying barrier physiology, allergen penetration, atopic-march initiation, and candidate therapeutics. **Resources:** MGI (mouse Flg).

Mechanistic Model / Interpretation



The unifying interpretation is that **a single structural deficiency — loss of filaggrin — produces two coupled outputs**: (1) a *structural/hydration* failure that generates the visible IV phenotype, and (2) a *permeability* failure that opens an epicutaneous route for allergic sensitization. The first output is constitutive; the second is conditional, requiring environmental allergen/irritant exposure and inflammation (Findings 11–12). This explains why IV and atopic disease co-segregate yet are dissociable, why biallelic carriers are more severely affected (dose-dependent protein loss, Finding 2), and why the disease fluctuates with ambient humidity (Finding 4). Therapeutically, it explains why **barrier restoration** (emollients, NMF/ceramide replacement, keratolysis) is both the mainstay symptomatic treatment and a rational preventive strategy against the atopic march.

Evidence Base

PMID	Finding(s) supported	Contribution
17573887	F1, F2	R501X/2282del4 causal; semidominant inheritance
22158554	F1, F6	FLG LOF causes IV, "most common disorder of keratinization"
27519469	F2	Biallelic FLG → severe IV; gene dosage
30021537	F2	STS modifier exacerbates IV
23301728	F3, F5	Ancestry-specific FLG frequencies; core clinical features
19349982	F3, F6, F10	European cumulative freq ~9%; flaky-tail mouse; allergen priming
21173567	F3	~40 population-specific LOF alleles
33462753	F4	Filaggrin keratin aggregation + NMF catabolism enzymes
28242341	F4	Low humidity → filaggrin deimination/breakdown
38841231	F5	Reduced/absent granular layer histologic hallmark
34608691	F5	Palmar hyperlinearity diagnostic accuracy for FLG
33894197	F6, F10	FLG loss compromises barrier; FLG-deficient mouse phenotype
21576945	F6	Filaggrin hypothesis; AD/asthma association
32115871	F7, F8, F9	Hydration/acitretin mainstay; clinical diagnosis; first-year onset
36159354	F7, F8, F9	Glycolic acid peel ~90%; differentials; ~37–50% eczema
42707485	F7	Gene therapy emerging; management "largely palliative"
36165597	F8	Acquired ichthyosis mimics IV histologically
39381601	F8	Upfront WGS for primary atopic/barrier disorders
42001132	F9	Lifelong QoL impact (n=86 IV)
42490938	F9	Psychosocial burden in pediatric skin disease
40042058	F10	Reduced filaggrin in canine atopic skin
23343419	F11	GxE: FLG sensitization conditional on dermatitis (OR 5.71)

PMID	Finding(s) supported	Contribution
30868611	F11	R501X → polysensitivity
26872425	F11	FLG null → persistent hand eczema (OR 3.1)
32249942	F12	Dual-allergen hypothesis; FLG/Langerhans/ILC2/IL-33/TSLP
29676818	F12	Atopic march = self-sustained epithelial dysfunction
28899689	\$6	IV has fewer transcriptomic changes than AD
38336337	\$4, \$10	Nanopore FLG phasing/CNV/methylation

Note on evidence quality: Causality, mechanism, and genetics are supported by strong **human genetic and biochemical** evidence plus **mouse-model** validation. Epidemiology figures are ancestry-specific and derive from carrier-frequency and cohort studies. Treatment evidence is largely observational/case-series (no cure); gene therapy is preclinical/early. One citation

([P 33894197](#))

was flagged as a title-based mismatch in the knowledge state but its barrier/atopy content aligns with Findings 6 and 10.

Limitations and Knowledge Gaps

1. **Prevalence precision.** Exact clinical prevalence of IV is uncertain and ascertainment-dependent; carrier frequencies are better characterized than diagnosed-case prevalence.
2. **Genotype–phenotype quantitation.** The semidominant dose effect is established qualitatively (Finding 2), but graded severity scores across mono- vs biallelic genotypes in large cohorts are limited.
3. **Modifier landscape.** Beyond STS, the full spectrum of severity modifiers (other EDC genes, lipid-processing genes) is incompletely mapped.
4. **Model limitations.** FLG-deficient mice reproduce the barrier/sensitization axis but **not spontaneous dermatitis** (Finding 10), limiting inflammatory-mechanism studies.

5. **Epigenetics.** Disease-specific methylation/chromatin signatures in IV are not established; nanopore data are preliminary.
6. **Therapeutics.** No disease-modifying therapy; gene/RNA-directed correction of FLG is unproven in humans.
7. **Diverse ancestry.** FLG mutation spectra and IV phenotypes in African and admixed populations remain under-characterized.
8. **Prevention evidence.** Whether early emollient therapy durably prevents the atopic march in FLG carriers remains debated and trial-dependent.

Proposed Follow-up Experiments / Actions

1. **Genotype-stratified natural-history cohort** quantifying IV severity (validated scale) by FLG allele count and specific alleles, to formalize the dose-response and identify modifiers via GWAS/WES.
 2. **Ancestry-broadening sequencing** (including African and admixed populations) of FLG to define region-specific LOF spectra and refine global epidemiology.
 3. **FLG correction proof-of-concept** in human skin equivalents/organoids using CRISPR base-editing or profilaggrin re-expression, measuring NMF restoration and barrier permeability.
 4. **Prospective emollient-prevention trial** in FLG-carrier infants with allergic-sensitization and AD-incidence endpoints, directly testing the Finding-12 prevention hypothesis.
 5. **Multi-omic IV vs AD contrast** (transcriptomics + lipidomics + NMF metabolomics) on matched nonlesional skin to delineate the structural-only IV signature from the inflammatory AD signature (building on [P 28899689](#)).
 6. **Humidity-intervention study** testing whether controlled ambient humidification measurably reduces PAD-mediated flaggrin breakdown and clinical scaling (translating [P 28242341](#)).
 7. **Standardized NMF/palmar-hyperlinearity biomarker validation** as a low-cost proxy for FLG genotype in clinical screening.
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Consensus Answer

Ichthyosis vulgaris (MONDO:0007810; OMIM 146700) is the most common inherited disorder of keratinization, caused by autosomal semidominant loss-of-function (null) mutations in the filaggrin gene FLG (1q21.3; founder alleles R501X and 2282del4), with biallelic carriers showing more severe disease. Filaggrin deficiency impairs keratin-filament aggregation and abolishes filaggrin-derived natural moisturizing factor, producing xerosis, fine scaling with flexural sparing, keratosis pilaris and palmoplantar hyperlinearity (histologic hallmark: reduced/absent granular layer), while the same barrier defect drives atopic dermatitis, asthma and contact sensitization. It is a chronic, lifelong, non-life-threatening condition managed symptomatically with hydration, emollients and keratolytics (acitretin for severe cases); there is no cure.

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